Herded or Anchored? Conformity at ICLR during the Foundation-Model Wave

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Abstract

Machine learning, NLP included, is said to herd: researchers chase what is hot, and review rewards it. We test this account on 43,480 ICLR submissions (2017–2026), rejected and withdrawn papers included, with hypotheses registered before outcomes were read and tested again on 11,099 held-out papers. We separate two routes to conformity: imitation takes its cue from what others do now, continuity from an accumulated record. The field converged: topics with more foundation-model papers the year before had fewer new-architecture papers. Hindsight inflates emergence: on the same papers, a topic map fitted on all years finds 29 emerging topics, one fitted on 2017–2020 only 8. Among topic newcomers (no author had submitted there), where imitation theory expects herding most, our registered prediction of follow-on work failed in the opposite direction: their papers more often propose methods and less often benchmarks, datasets or evaluations, and the reversal replicated. Review shows no detectable premium for crowded topics. In post hoc tests supported on held-out papers, acceptance is lower far from the team's earlier work and higher with a topic insider on the team. At ICLR, the conformity we can detect points to a record, not to the crowd. All findings are associations.

1 Introduction

Machine learning is said to herd. Between 2017 and 2026, yearly submissions in our discovery corpus of the International Conference on Learning Representations (ICLR) grew 40-fold, from 384 to 15,411. The share building on a foundation model (FM) rose from 0.3% to 54.4%, and the share proposing a new architecture fell from 24.5% to 7.5%. Keywords rose and fell (Figure 1): generative adversarial networks fell from 8.1% of submissions in 2018, and large language models reached 16.7% in 2026. The wave is also NLP's, whose researchers increasingly publish, and debut, at ML venues (Jurgens, 2026). Critics of machine-learning scholarship point to incentives that favour short-term success (Lipton and Steinhardt, 2019) and to convergence on what suits the field's hardware, benchmarks and foundation models (Hooker, 2021; Dehghani et al., 2021; Bommasani et al., 2021). In cascade and fashion theories, people take their cue from others (Bikhchandani et al., 1992; Banerjee, 1992; Abrahamson, 1991). Read as herding, the waves make three claims: they are fashions, imitators carry them, and following pays at review.

Convergence, however, does not require copying. Conformity, a pull towards an existing reference point that need not make papers identical, has two routes that the herding account runs together. *Imitation* takes its reference point from what others are doing, or are rewarded for, now: popularity, momentum, recent acceptance. *Continuity* takes it from an accumulated record: the team's own past work, or a topic's past work carried by a team member who has submitted in it. Weisberg and Muldoon (2009) model "followers", drawn to approaches others have already explored, apart from agents who ignore others' findings and from "mavericks". Later work models herding in science as risk-avoiding clustering (Weisberg, 2013), which either route could produce. The route matters: imitation can discard private information and leave a fragile consensus (Banerjee, 1992; Bikhchandani et al., 1992), while continuity keeps work near existing competence, with short-run gains and long-run costs (March, 1991).

We call work that conforms by continuity *anchored*.[1] Our thesis is that the conformity we can detect at ICLR is anchored to a record, not herded

[1]We use 'anchored' for work that conforms by continuity, a position relative to a record; it is not the anchoring-and-adjustment heuristic (Tversky and Kahneman, 1974), the sense in which writing on herding in science uses the word (Baddeley, 2015).

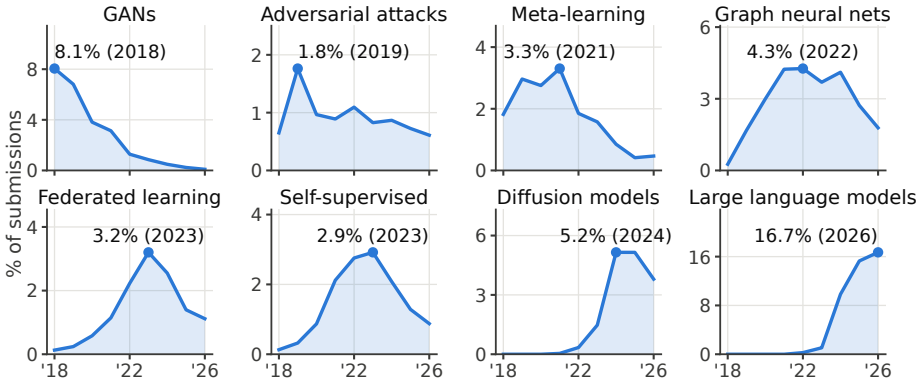


Figure 1: Research waves at ICLR: share of discovery submissions listing each author keyword, 2018–2026; dots mark the peak. The eight keywords were chosen for display after their curves were known. Across the 68 most frequent keywords, 17 peaked in 2019–2024 with shares of at most half the peak both before it and in 2026 (discovery shares; these are the waves of Section 6). Outcome-free.

by the crowd. The field did converge on a shared package (A), as both routes predict. But where the routes part, at entry to a topic and at review, the evidence points away from imitation. We expected a herd too: before any outcome was read, we registered that topic newcomers would write fewer method papers and more benchmark and evaluation papers. The prediction failed in the opposite direction.

Prior work relates ICLR acceptance to prestige, process and research area (Tran et al., 2020; Ibrahim et al., 2026; Jung et al., 2025; Wang et al., 2023; Xu et al., 2026) and describes which ICLR papers start topics, who enters LLM research and how far researchers stay from their own past work (Huang, 2026; Movva et al., 2024; Bao and Li, 2025). Moving far from one's own record goes with fewer citations and lower publication success (Hill et al., 2025), and field histories read trends from topic models fitted on the whole period (Hall et al., 2008; Anderson et al., 2012). To our knowledge, among the works we checked, no study tests all three claims of the herding account within one submission population, rejected and withdrawn papers included, with registered hypotheses tested again on held-out papers, as Ibrahim et al. (2026) did for a prestige gap.

We test the account on 43,480 ICLR submissions, 2017–2026. Hypotheses were registered before any full-corpus label existed, tested once in a frozen run, and tested again on 11,099 *held-out* papers: a pre-registered out-of-sample replication, not a certified untouched holdout. Every result carries an evidence tier: **A** registered and replicated; **B** post hoc, then supported by a declared held-out test; **C** registered, inconclusive or not replicated; **D** discovery only; **E** descriptive. All results are associations.

For theories of herding and fashion, we contribute a registered test of imitation at entry, where its prediction is sharpest, read post hoc against continuity, a route long separated from it in theory (Levitt and March, 1988; Weisberg and Muldoon, 2009). The reversal replicated (A), and teams new to ICLR come later to the package (E). For the science of science, acceptance goes with a paper's distance from its own authors' record (B), in line with the pivot penalty (Hill et al., 2025), while crowded topics carry no detectable premium (C). For computational histories of NLP and ML, hindsight inflates emergence: on the same papers, an all-years topic map finds 29 emerging topics and an early map 8 (E). At ICLR, the conformity we can detect points to a record, not to the crowd.

2 Two Routes to Conformity

Conformity alone does not reveal its reference point. Imitation makes teams alike; continuity keeps each team, and so the field, near its past.

Imitation. We measure frequency imitation, through crowding, hot topics, rising keywords and FM adoption, and outcome imitation, through topics with high past acceptance, but not trait imitation (Haunschild and Miner, 1997). Uncertainty breeds mimicry (DiMaggio and Powell, 1983) and cascades (Bikhchandani et al., 1992), so topic entrants are predicted to imitate most. They would flock to hot topics, take up a rising package fast, and build on what others proposed by evaluating or benchmarking it: the follow-on work we registered.

Reputational herding pays when evaluators judge people relative to their peers (Scharfstein and Stein, 1990), and bandwagons can spread without reward (Abrahamson and Rosenkopf, 1993).

Continuity. Search stays near existing competence (March, 1991; Stuart and Podolny, 1996), and absorptive capacity depends on prior related knowledge (Cohen and Levinthal, 1990). Teams without a record would thus take up a new package slowly, against the cascade prediction. At entry, continuity predicts that incumbents consolidate the topic's record. Producer-side studies find that fresh teams and younger researchers try new ideas (Zeng et al., 2021; Packalen and Bhattacharya, 2019); on such *outsider accounts*, entrants bring other work. In Bourdieu's field (1975), "the dominant are committed to conservation strategies" and journals "consecrate productions faithful to the principles of official science". Official science, like tradition (Foster et al., 2015), is a third reference point, others' accumulated work, which both routes approach, so distance from it cannot separate them. We measure continuity by where a paper's content sits relative to its team's record and, more weakly, by a topic insider on the team. An insider is a record possessed, which cumulative advantage (Merton, 1968) and, where authors are identifiable, visible status (Ibrahim et al., 2026) predict too.

Timing and the field level. Timing separates rewarded imitation of what is popular, under which late riders do as well as early ones, from priority, under which the premium falls as a wave matures (Merton, 1957; Strevens, 2003). A package bandwagon (Fujimura, 1988), lock-in (Arthur, 1989) and efficient choice (Sutton, 2019) all predict convergence on a package, so only entry and the gate can tell the routes apart.

Gap. To our knowledge, among the works we checked, imitation and continuity have not been tested against each other at entry, or side by side at the review gate. A recent review of topic choice names social context and path dependence as drivers and leaves entry costs open (Xiang et al., 2025). We use the Weisberg–Muldoon taxonomy, not its welfare conclusions, which are disputed (Alexander et al., 2015); our terms, adopted after the tests ran, label papers' positions, not motives.

3 Data and Design

The tests need every submission, past-only predictors and sealed outcomes.

Corpus. The corpus is our own dated acquisitions of official OpenReview records (Appendix A). Acceptance tests use 32,190 decided discovery papers, 40.1% of them accepted, and 8,185 decided held-out papers; withdrawn papers are left out.

Instruments. A *topic* is one of 100 spherical k-means clusters of SPECTER2 embeddings of title and abstract (Cohan et al., 2020; Singh et al., 2023), fitted on all years unless another map is named. An LLM with a fixed codebook gives each paper one primary contribution type. A *method paper* includes new architectures, a *measurement paper* is a dataset/benchmark or evaluation paper, and an *FM paper* uses or adapts a large pre-trained model. A blind second pass agreed at κ 0.93: consistency between LLM labellings, not human-level validity. In a *newcomer paper*, no author ID has an earlier ICLR submission, with any outcome, in the paper's topic (a topic-relative definition, as in Biliotti et al., 2024); otherwise it is an *incumbent paper*. A *first-time team* has no earlier ICLR submission at all; a *returning team* has one. A *returning topic newcomer* paper is a newcomer paper with an author who has one in another topic, and a *topic insider* is an author with one in the paper's topic.

Two distances. Both use the centred SPECTER2 space. Our closeness measure follows Bao and Li (2025), who use SPECTER2 similarity to a researcher's own earlier papers. $F_{\max}$ is the highest cosine between the paper and the mean of an author's earlier submissions, taken over *experienced authors*, those with an earlier ICLR submission since 2021 (exclusions in Appendix B). *Distance from earlier ICLR work* is one minus the mean cosine to the 10 nearest earlier-year submissions, as a percentile within topic $\times$ year.

Prospective measurement. All predictors use only the past. *Prior FM adoption* is the FM share of a topic's previous-year papers, *crowding* is the log of one plus the topic's earlier papers, and a *hot topic* is in the top 20% of prior growth. Code links leak the outcome into 17.6% of accepted but 0.5% of rejected 2023–2026 abstracts (Kaufman et al., 2012); T✓ in Table 1 marks estimates that held after we removed them.

#	Observable [code]	Predicted: imitation / continuity	Discovery [95% CI]	Held-out [95% CI]; MDE80 or power	Tier	Reading
(i) The field: are the waves fashions?						
1	Emerging topics: all-years vs 2017–2020 map [hindsight vs P]	· / ·	29 (40.9%) vs 8 (22.3%)	–	E	hindsight inflates emergence
2	Prior-year FM adoption 0→100% → new-architecture paper [P, H✓]	− / − (lock-in)	−7.98 [−12.14, −3.82]	−11.0 [−18.1, −3.9]; MDE80 12.0[a]	A	convergence; does not discriminate
3	Twins per paper per 1,000 submissions, 2019→2026 [P]	+ / ·	0.32→0.19	–	E	volume, not piling on
(ii) Entry: what topic newcomers write						
4	Newcomer → method paper (base 75.0%) [P, H✓]	− (registered) / +	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]; MDE80 2.8	A (rev.)	against imitation[h]
5	Newcomer → measurement paper (base 7.7%) [P, H✓]	+ (registered) / −	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]; MDE80 2.4	A (rev.)	against imitation; partly team size[g]
6	Newcomer → distance from earlier ICLR work [P]	− / + (incumbents' own work)	+5.90 [+4.95, +6.85]; own work excluded +4.33 [+3.44, +5.22]	–	E	against imitation[i]
7	First-time vs returning team → Emerging keyword; first-time team vs returning topic newcomer → FM paper at 5–15 / 15–30 / 30–50% adoption, 2023–2026 [P]	+ / − (absorptive capacity)	−3.74 [−5.14, −2.35]; −4.58 / −8.96 / −14.71	–	E	later onto trend and package
8	Hot topic → incumbent share (flocking) [P, H×]	− / ·	−0.73 [−2.68, +1.22]	not registered	C	inconclusive
(iii) The gate: following and timing						
9	Crowding, per log unit → acceptance [P, H×]	+ if rewarded (registered: −) / ·	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]; MDE80 4.8	C	no detectable premium[b]
10	Topic edge 2018–2022 → edge 2024–2026 [P]	$b > 0$ if outcome imitation pays / ·	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	B	real but regressing
11	Emerging minus Mainstream keyword [P]; after vs up to a keyword's peak, 17 waves [R]	+ / + (priority); 0 or − / −	+2.67 [+0.76, +4.58]; −5.7 [−8.5, −2.9]	+3.25 [−0.18, +6.69]; −5.2 [−11.1, +0.7] (MDE80 7.4)	B, weak	momentum vs priority open; shape leans to priority (D)[c]
12	Federated-learning keyword gap [R]	· / ·	−11.3 [−17.9, −4.8]	−4.9 [−15.4, +5.7]; MDE80 14.5	D/C	not confirmed[d]
(iv) The gate: distance from the field's and from the team's own record						
13	Field distance: least-distant vs middle fifth [P, T✓]	+ if rewarded (declared: −) / + (field record)	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]; power 0.47	C	undetermined; does not discriminate[e]
14	Far jump from own record: $F_{\max}$ p10 − p50 [P, T✓]	· / −	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]; MDE80 4.9	B (prim.)	for continuity[f]
15	Same author, per +0.1 cosine closer [P, T✓]	· / +	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	B (sec.)	for continuity
16	One topic insider vs none [P]	· / +	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]; MDE80 ≈4.5	B (sec.)	status-compatible

[a] Underpowered at the discovery size: direction, not size. [b] Registered H3a predicted a penalty (−); one-sided held-out p 0.21; the hindsight requirement fails held-out (early map on 2021–2026 papers +1.44). Penalties below about 5 pp per log unit cannot be told from zero; premiums above +0.32 / +2.19 pp per log unit lie outside the intervals. [c] 'Early' means early in a keyword's life, not in the submission window (cf. Jung et al., 2025). Imitation predicts + for Emerging if momentum is rewarded, and after the peak 0 (popularity) or − (momentum). The Emerging contrast's two-sided interval includes 0 (MDE80 ≈4.3); old rising keywords +3.9 (nominal). Discovery wave estimate descriptive, not a test; wave set retrospective; held-out −1.7 (p 0.33) without the six keywords inspected first. In discovery the gap falls before the peak, and rising against flat years of the same keyword differ by +0.9 [−0.8, +2.7]. [d] Held-out trend of opposite sign (+3.2 pp per year). [e] Uninformative. Rows 13–16 differ in population and unit: compare replication status, not size. [f] Newcomer control −5.0; link-removed −5.39; early map −5.2; correlation with newcomer flag −0.57. [g] With log team size: −0.54 [−1.11, +0.04] in discovery, −0.98 held-out; row 4 unchanged (+3.92 in discovery, +3.64 held-out); supplementary to A. [h] Rows 4–5: all 10 counted variants agree, and the t-statistics (6.7, −4.4) dwarf the size-failure inflation factors (1.31, 1.22). [i] Raw scale: +1.35 [+1.16, +1.55] hundredths of cosine distance, 0.22 of a within-cell standard deviation.

Table 1: The herding account tested at three points: each observable and the sign predicted by imitation and by continuity (in parentheses: its mechanism or reference point, or a named alternative: lock-in, priority; ·: none). Percentage points unless the row states otherwise (row 1: counts; row 3: twins per paper per 1,000 submissions; row 6: percentile points; row 10: slope b), 95% intervals, within topic and year (keyword rows and row 10: against the same year's other papers). Held-out tests are one-sided and confirm direction, not size, so a supported test can have an interval that includes 0. Codes: P only earlier data; H✓/H× registered hindsight variants pass/fail; R retrospective by construction; T✓ held after link removal. Tiers as in Section 1; rev.: reversed registered prediction; prim./sec.: declared primary/secondary held-out test (secondary tests, six in all, among them rows 10, 11 (Emerging), 15 and 16, were one-sided and unadjusted, so one false positive would not be surprising); D/C: discovery-only gap whose registered held-out test was not supported. Every test, and further notes to rows 6, 7, 10 and 16, in Appendix D.

Protocol. A registered hypothesis was supported if it had the predicted sign, Holm-adjusted $p < 0.05$ in its family (Holm, 1979), and that sign in 75% of counted variants and in both hindsight variants. These are a year-balanced map and the 2017–2020 map applied to 2021–2026 papers. Outcomes stayed sealed until one frozen run (Nosek et al., 2018; van den Akker et al., 2021); post hoc families were declared before they were run, and held-out tests were registered beforehand and run once. Models are linear probability models with topic and year fixed effects and topic-clustered errors. Under a null that cuts across topics, the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%, which a wild cluster bootstrap (Cameron et al., 2008) did not fix (Appendix H). A test not

supported and below its 80%-power minimum detectable effect (MDE80) is *uninformative*.

4 The Field: One Package, Less Emergence than Hindsight Shows

The herding account's first claim is that the waves are fashions: rises that will fall. We ask what survives a measurement that cannot see the future.

Hindsight inflates emergence. Computational histories fit one topic model on the whole archive and read it forward (Hall et al., 2008; Anderson et al., 2012; Klinger et al., 2020), and dynamic topic models and BERTopic also fit all years at once (Blei and Lafferty, 2006; Grootendorst, 2022). Some attributes of emergence can be judged only after the fact (Rotolo et al., 2015), static models can miss small emerging topics (Li et al., 2025), and a prospective signal has been tested (Khanbayov and Kurban, 2026). An all-years map leaks the future into the past, like temporal leakage in prediction (Kapoor and Narayanan, 2023) and hindsight in judgement (Fischhoff, 1975). *Emerging topics* are those whose 2024–2026 share is at least three times their 2017–2020 share. On the same papers and pipeline, their number depends on the fitting years: 8 on the 2017–2020 map, rising with later fitting years to 29 on all years (Figure 2a; E). With 69% of papers from 2024–2026, the all-years map is a present-day map. The share of 2024–2026 papers in emerging topics, 22.3% on the 2017–2020 map and 40.9% on the all-years map (20.9–42.2% across the six maps), is a bracket. The early map cannot represent topics born after 2020, and the maps differ in fitting corpus as well as window. A registered trend conclusion also changes with the map (Appendix C).

What survives: a package. What survives every map is a package. The FM share rose every year from 2018 (Figure 2b). Of the 68 most frequent keywords, 17 instead rose and fell (E). Registered H1d held: where more of a topic's previous-year papers built on FMs, fewer of its papers propose new architectures. The estimate is −7.98 pp [−12.14, −3.82] for adoption from 0 to 100% in discovery and −11.0 [−18.1, −3.9] held-out (A, direction only). That is about a third fewer at 50% adoption than at 10%, on a base near 9%. No detectable shift towards empirical or evaluation papers goes with it (−0.63 [−3.64, +2.38]; C), and reverse causality is possible. The pattern is consistent with accounts

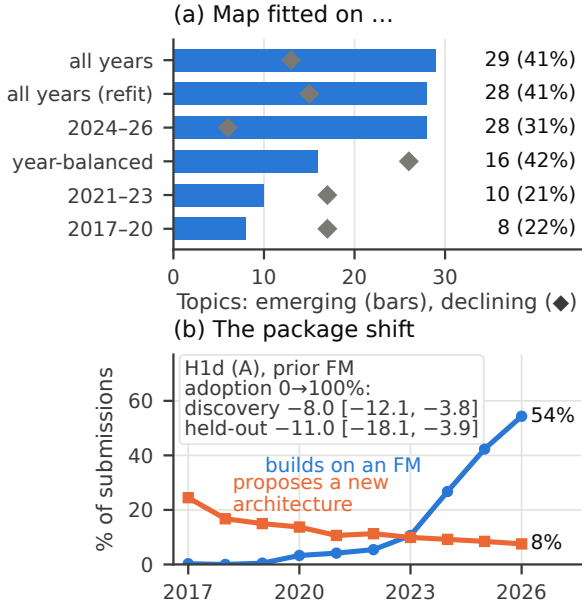


Figure 2: (a) Emerging (bars) and declining (grey diamonds) topics on $k = 100$ maps fitted on different years, same 43,480 papers; right-hand labels: emerging count (share of 2024–2026 papers in emerging topics). All years (refit): the all-years map refitted with another seed (Table 6). Topic size 'predicts' two-year growth only on the all-years map (R^2 0.28 vs −0.03; no paired test). (b) Share of submissions building on an FM (blue circles) and proposing a new architecture (orange squares); LLM labels. Inset: registered H1d (A; held-out confirms direction, not size). No acceptance data.

of NLP's paradigm shifts (Gururaja et al., 2023; Saphra et al., 2024; Pramanick et al., 2023).

What the field level cannot decide. Papers with an independent *same-year twin*, at cosine ≥ 0.70, grew with volume but not per possible pair (E; Table 1). Imitation-like spread is plainest in surface form: *branded titles*, a short coined name and a colon, rose about fourfold (E*). An earlier exploratory analysis, not re-run, found them accepted at the submitted rate (ratio 1.02; D*). The first claim thus holds only in part: keywords rose and fell, but emergence depends on the map, and the largest shift, the package, rose every year. It fits both routes, and lock-in (Hooker, 2021), so the field level cannot tell imitation from continuity; entry and the gate must.

5 Entry: Newcomers Propose Rather than Follow

The herding account's second claim is that imitators carry the waves: newcomers to a topic follow where others have led. A randomised trial at ICML

detected no herding among reviewers in discussion (Stelmakh et al., 2023), though co-reviewers' ratings predict rebuttal-phase score changes at ICLR (Kargaran et al., 2025). We ask about authors entering a topic, where imitation theory expects herding most and continuity the opposite.

The registered prediction reversed. Newcomer papers are 72.9% of 42,325 papers. Our registered herding prediction, that they would be follow-on work, failed in the opposite direction, and the reversal replicated (A; Figure 3a). Newcomer papers are more often method papers, +3.71 pp [+2.61, +4.80] in discovery and +3.60 [+1.48, +5.71] held-out. They are less often measurement papers, −1.70 [−2.46, −0.93] in discovery and −2.10 [−3.35, −0.85] held-out (direction only). All counted variants agree (Table 1, note h), and the reversed hypotheses were registered before any held-out data.

What the reversal means. A method paper is a contribution type, not a measure of novelty; an adapter on an LLM counts as method. As three quarters of ICLR papers are method papers, newcomers could also be imitating the venue's dominant genre, a reading our registered prediction did not encode. Against it, first-time teams come later to its dominant package, and newcomer papers sit farther from earlier work. Incumbent papers carry more of a topic's measurement work, the consolidation continuity expects, but much of that gap is team size, unlike the method gap (Table 1, note g).

Farther from earlier work. Newcomer papers also sit farther from earlier ICLR work, in an unregistered comparison added during writing (E). The gap is +5.90 percentile points [+4.95, +6.85] within topic × year, +5.24 [+4.16, +6.31] among method papers alone, and +4.33 [+3.44, +5.22] with the authors' own earlier papers excluded from the reference set. About a quarter of the gap is thus incumbents' closeness to their own work, itself continuity among producers; the rest is distance from the field's earlier work. The gap is modest on the raw scale (Table 1, note i), and distance in SPECTER2 space is not originality. But the follow-on prediction expects newcomer papers to sit closer to earlier work, and they do not. They also less often have a same-year twin, −3.86 [−4.98, −2.74] (E; Figure 3a).

Later onto the package and the trend. First-time teams come later to both the package and the

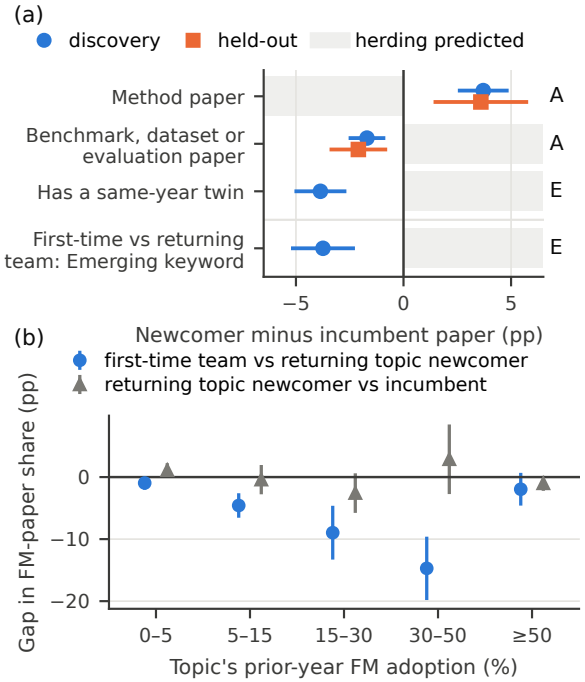


Figure 3: (a) Newcomer minus incumbent papers within topic and year (pp): discovery (blue circles), held-out (orange squares), 95% intervals; grey band: the side the herding prediction expected; twin row: class, FM and team-size controls. Below the line: first-time minus returning teams. A method paper is a contribution type, not a measure of novelty. (b) FM-paper share gap by the topic's prior-year FM adoption, 2023–2026, class and team-size controls: first-time team vs returning topic newcomer (blue circles); returning topic newcomer vs incumbent (grey triangles); 95% intervals. Outcome-free.

trend (E; Figure 3a–b). Among 2023–2026 submissions in topics at 5–50% prior adoption, the FM-paper gap between first-time teams and returning topic newcomers runs from −4.58 pp [−6.55, −2.60] to −14.71 [−19.82, −9.61]. Returning topic newcomers do not differ detectably from incumbents. First-time teams also list *Emerging* keywords, young and rising, less often than returning teams: −3.74 [−5.14, −2.35]. The package travels with an ICLR record, the order absorptive capacity predicts (Cohen and Levinthal, 1990) and the reverse of younger researchers' pull to new ideas (Packalen and Bhattacharya, 2019) if first-time teams are younger. The compute and access a record brings predict the same order.

What stays open. The registered flocking test is inconclusive (C): the incumbent share in hot versus other topics differs by −0.73 [−2.68, +1.22], pointing the imitation way. Whether entrants crowd into

hot topics thus stays open. The method, measurement and distance gaps speak for continuity only through incumbents. The newcomer side fits outsider accounts as well, since entering a topic is itself a move away from a team's ICLR record: $F_{\max}$ correlates -0.57 with the newcomer flag. The FM contrast between topic newcomers with and without an ICLR record does separate continuity from outsider accounts. At entry, then, every detectable difference runs against imitation: newcomers write less follow-on work, not more, and the package travels with an ICLR record.

6 The Gate: What Acceptance Goes With

The herding account's third claim is that following pays at review. The negative half of our thesis rests on a registered test (C), the positive half on post hoc tests supported by declared held-out tests (B).

Crowd, package and past winners. The registered crowding test, which predicted a penalty, is uninformative: -1.68 pp per log unit [-3.68, $+0.32$] in discovery and -1.51 [-5.20, $+2.19$] held-out, against an MDE80 of 4.8 (C). Crowded topics carry no detectable premium (Table 1, note b). FM papers, the package itself, are accepted more often, +4.62 pp [+2.79, +6.44] (D). That premium does not detectably change with the topic's prior adoption, +0.19 [-0.96, +1.34] per logit unit (D): late adopters fare as early ones do, which fits rewarded imitation and efficient choice alike. Topics differ in acceptance at ICLR (Wang et al., 2023), even at equal reviewer scores (Xu et al., 2026); we find these edges real but regressing. A *topic edge*, acceptance minus that of all other papers of the same year, in 2018–2022 predicts the 2024–2026 edge with slope b of 0.38 [0.18, 0.58] in discovery and 0.60 [0.32, 0.88] held-out. With $0 < b < 1$ (B), a favoured topic keeps part of its edge, not all of it.

Continuity: closeness to the team's own record. Holding the author fixed, a paper 0.1 closer in cosine to that author's earlier work goes with +0.55 pp [+0.21, +0.90] higher acceptance in discovery and +1.06 [+0.23, +1.89] held-out (B, secondary). Coauthor composition may explain half or more of this. The declared primary test compares teams that all have a record. Among 20,430 decided 2023–2026 papers with an experienced author, the *far jump*, acceptance at the 10th percentile of $F_{\max}$ minus that at the median, is -5.44 [-7.10, -3.78] in discovery and -6.09 [-9.32, -2.86] held-out, with power 0.89 (B). Adjusted acceptance rises with closeness up to the fourth fifth (D; Figure 4). Larger pivots are less often published (Hill et al., 2025); we observe the decision itself, on every submission, under double-blind review, with a declared held-out test. Unlike distance from an evaluator's expertise (Boudreau et al., 2016) or reference-based novelty (Teplitskiy et al., 2022), ours runs from a submission to its own authors' record.

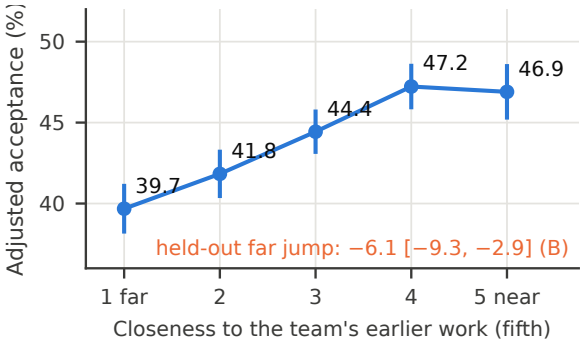


Figure 4: Adjusted acceptance by fifth of closeness to the team's earlier ICLR work, far to near, 95% intervals (discovery, 2023–2026, papers with an experienced author; D); annotation: declared held-out far-jump test (B).

A topic insider: a record possessed. At fixed team size and number of experienced authors, one topic insider on the team, against none, goes with +3.42 pp [+1.18, +5.66] in discovery and +3.88 [+0.28, +7.47] held-out (B, secondary, direction only). A status reading (Merton, 1968; Ibrahim et al., 2026) explains this as well as continuity does, and a discovery-only contrast leans that way. Per head, an insider goes with only +0.58 pp more than an experienced author from another topic (p 0.16; D).[2] Chaperoned entry into top journals (Sekara et al., 2018) and, on the producer side, expertise that offsets a newcomer disadvantage (Biliotti et al., 2024) are parallels.

Two records in one space. In the same space, closeness to the field's earlier papers is undetermined (Appendix D, Figure 5). The least-distant fifth against the middle fifth is -3.27 [-4.93, -1.60] in discovery, the declared direction, but -0.13 [-3.50, +3.25] held-out, with power 0.47 (C). The within-author slope for field distance was not supported either (C). Only the own-record associations held on held-out papers.

[2] Newcomer papers are accepted less often, -10.2 pp [-11.8, -8.7] (D), but that gap mostly reflects who writes (Appendix F).

	Producers (what is written)	Gate (acceptance)
Imitation (others now)	Follow-on prediction reversed (A); newcomers farther from earlier work (E); first-time teams later onto FMs and rising keywords (E); fewer newcomer twins (E). Flocking inconclusive (C). Spread plainest in title form (E^*).	No detectable crowding premium (C); branded titles at the submitted rate (D^*); late riding not favoured (B, weak). Fit imitation and a rival alike: rising keywords weakly favoured (B, weak); topic edges partly kept (B); FM premium not fading (D).
Continuity (a record)	Incumbents write more measurement papers (A; much of it team size); the package travels with an ICLR record (E); incumbents closer to their own earlier work (E).	Lower far from the team's record (B); higher for the same author's closer papers (B); higher with a topic insider (B; a record possessed, status-compatible).

Table 2: Where conformity shows, by route and site. Tiers in parentheses. Cells summarise Table 1 and Sections 4–6; no new estimates. Field distance (Table 1, row 13), predicted by both routes, is undetermined (C).

Timing: early, weakly. Emerging keywords go with higher acceptance than *Mainstream* ones, which are old and flat: +2.67 pp [+0.76, +4.58] in discovery and +3.25 [−0.18, +6.69] held-out, a two-sided interval that includes 0 (B, weak). Old rising keywords do about as well (+3.9, nominal; D), so this fits a premium for what is rising as well as priority. *Waves* are the 17 keywords whose share peaked in 2019–2024. Acceptance after the peak, against up to it, is −5.7 [−8.5, −2.9] in discovery, not a test, and −5.2 [−11.1, +0.7] held-out (B, weak). The wave set is retrospective, and the held-out estimate leans on the six keywords inspected first (without them, −1.7, p 0.33). A premium for what is rising would also fall after a peak, but the discovery gap falls before it, which leans towards priority (Merton, 1957; Strevens, 2003) (D). Neither test shows the flat premium that rewarded imitation of what is popular predicts.

What acceptance goes with. Of the post hoc patterns taken to held-out papers, the three referenced to a record and the topic-edge regression held. The timing tests held weakly, and field distance and the federated-learning gap (Table 1, row 12) were not confirmed (Appendix D). Every association that held is referenced to a record or fits imitation and a rival alike. At the gate, acceptance goes with a team's record, not detectably with the crowd.

7 Discussion: Where Conformity Lives

What the evidence supports. The conformity we can detect at ICLR points to continuity rather than imitation, though three gate patterns fit rewarded imitation and a rival alike (Table 2). Conformity here is a pull towards a record, not identical work: incumbents' papers sit closer to their own earlier work (E), and acceptance goes with closeness to a team's record (B). Weisberg's risk-avoiding clustering (Weisberg, 2013) appears here as continuity. The consecration of "official science" that Bourdieu describes (Bourdieu, 1975) we detect only for a team's own line of work.

What remains open. Expertise, evaluator conservatism, visible status, cumulative advantage and selection into teams cannot be separated here (Li, 2017; Merton, 1968; Ibrahim et al., 2026). Double-blind review rules none of them out, since preprints can identify authors (Bharadhwaj et al., 2020). The within-author and far-jump tests add that acceptance goes with how close the content stays to a record, not only with having one, though close work may itself reveal its authors. Whether entrants' distant work is taken up less, as in the diversity–innovation paradox (Hofstra et al., 2020), our data cannot say. In the previous shift, researchers who stayed close to their own record had lower later impact (Bao and Li, 2025). Acceptance is a short-run outcome, and acceptance and later citation need not go with the same work (cf. March, 1991; Rzhetsky et al., 2015), which we cannot test.

Why the distinction matters. A field that looks herded may conform by continuity instead; the routes carry different risks: a fragile consensus for imitation, long-run costs for continuity (Bikhchandani et al., 1992; March, 1991). At ICLR the substrate converged (A), consistent with homogenisation (Bommasani et al., 2021), but no herding of the testable kinds was detected at entry. For histories of NLP and ML (Hall et al., 2008; Anderson et al., 2012), an early refit and lagged predictors show cheaply which trend claims survive hindsight.

8 Conclusion

We tested the herding account of ICLR's FM wave at three points, separating imitation from continuity. The field converged on a package, with less emergence than hindsight shows; newcomer papers reversed our registered herding prediction (A); crowded topics carry no detectable premium (C); and acceptance goes with a paper's position relative to its team's record (B). At ICLR, the conformity we can detect points to a record, not to the crowd: one venue, associations only, every claim tiered.

Limitations

All results are associations at one venue; none is a causal effect of a topic, team or contribution type. The held-out set is a pre-registered out-of-sample replication, not a certified untouched holdout: it shares conference-years, reviewer pools and authors with the discovery papers, so it guards against analyst overfitting but not against year- or reviewer-specific patterns, and supported tests confirm direction, not size. The continuity results were found post hoc; their held-out tests were declared in advance, but two of the three are among six secondary tests run one-sided and unadjusted. The imitation–continuity frame was drawn up after the registered and declared tests had run: it organises those tests rather than generating them, and the continuity predictions in Table 1 were written afterwards. Under one simulated null the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%. Contribution and foundation-model labels are LLM labels checked against LLM labels, and abstracts are partly camera-ready; links were removed and the text re-embedded for discovery papers only. Entry status counts ICLR submissions only, and the 2021 switch from e-mail to profile identifiers hides earlier histories, which pulls entry contrasts towards zero. Withdrawn papers are left out of acceptance tests. Whether entrants crowd into hot topics is unsettled (the flocking test is inconclusive), and the absence of a premium does not rule out unrewarded bandwagons. No declared held-out gate test pits imitation against continuity on the same unit; in discovery, the far-jump and insider contrasts in hot versus other topics are inconclusive (-0.35 [-3.13, $+2.42$] and -0.8 [-4.9, $+3.2$]; D). Work that continues a known line may itself reveal its authors, so even the within-author and far-jump associations do not exclude visible status. The insider and far-jump associations could be separated from it by asking whether they concentrate where a preprint identifies the authors, as the prestige gap at ICLR does (Ibrahim et al., 2026); we did not run that test. The hindsight comparison is a bracket, because the maps differ in fitting corpus as well as window. The content-distance comparison was added during writing and is unregistered; about a quarter of its gap reflects incumbents' closeness to their own earlier work (with that work excluded, the gap is +4.33 [+3.44, +5.22] percentile points), and distance in embedding space is not originality.

Ethics Statement

Author identifiers were used only as keys to compute entry status, closeness and the exclusion of a paper's own authors' earlier work; no author is named and all results are aggregates. The data are public OpenReview records, collected in our own dated acquisitions under the platform's terms. No review scores are used. Contribution labels were produced by an LLM (Claude Haiku) with a fixed codebook; agreement figures compare LLM labellings, not human judgement.

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A Data Provenance

The herding account is tested on one complete submission population, so that neither the entry contrasts nor the gate tests condition on acceptance. This appendix documents where the papers come from and which of them each test uses.

Year	Discovery	Held-out
2017	384	105
2018	770	212
2019	1,248	309
2020	2,070	500
2021	2,360	603
2022	2,649	704
2023	3,872	995
2024	5,765	1,445
2025	8,951	2,284
2026	15,411	3,942
All	43,480	11,099

Table 3: Papers with a usable abstract by year. Discovery: 2017–2020 4,472; 2021–2023 8,881; 2024–2026 30,127 (69.3%). Submissions grew 40.1-fold from 2017 to 2026. Outcome-free.

Acquisition. The corpus comes from our own dated acquisitions of official OpenReview records, reconciled into 55,938 submission instances. Titles, abstracts, author-declared keywords, primary areas, author lists and decisions all come from the selected official submission metadata; the keyword preprocessing is our own. The discovery export holds 43,516 submissions and 219,719 declared author-list positions (positions, not distinct people). The discovery corpus is the 43,480 submissions with a usable abstract of at least 30 words; 34 placeholders and 2 malformed abstracts are dropped. Abstracts are a September 2026 snapshot, so accepted papers more often carry camera-ready text: in 2023–2026, 17.6% of accepted and 0.5% of rejected abstracts contain a code link (Appendix F reports the link-removed re-run).

Splits. Table 3 gives papers by year. The held-out split is the 11,113 released former-holdout papers, of which 11,099 have an abstract, released on 2026-10-18. They were this study's confirmation split and had not been analysed before the held-out tests were registered. The source policy does not certify an untouched, independent holdout. The source does not document how papers were assigned to the split, and it excluded 1,309 pending family or link cases. We therefore call the held-out tests a pre-registered out-of-sample replication, not a certified untouched holdout. The two splits share conference-years, topic-year cells, reviewer pools and authors, so the held-out tests guard against analyst overfitting but not against year- or reviewer-specific patterns. The pooled corpus (54,579 papers) is used only for the exploratory re-run (Appendix D).

Analysis populations. Field analyses use all discovery papers (the registered field tests: 43,096 papers from 2018–2026; H1d and H1e 42,990). Entry analyses use the 42,325 papers from 2019–2026 with at least one author identifier (42,315 in 776 topic-year cells after singleton cells are dropped). Acceptance tests use the 32,190 ordinary submissions from 2018–2026 with a final accept or reject decision (40.1% accepted); the 9,986 withdrawn submissions are left out. Held-out populations are 10,978 papers (H1d), 10,718 (entry) and 8,185 decided papers from 2018–2026 (acceptance; 8,284 decided in all). The post hoc gate modules use decided 2023–2026 papers (Appendix F).

Existing collections, audited and not pooled. Re^2 (Zhang et al., 2025) was initially considered as the main dataset and audited: its main-conference ICLR coverage ends in 2023 and omits 2019, its conversations contain training-format placeholders, and its rebuttal train and test partitions share 475 paper identifiers. The audit motivated our own collection. A public GitHub archive of ICLR rebuttal records (`iclr-rebuttal-analysis`)[3] served as a separate layer for source comparison and identity reconciliation; it added no submissions and supplies no text. A public metadata archive (Paper Copilot)[4] was used for metadata comparisons and documents the lineage of reconstructed 2024–2025 review-score pairs; this study uses no review scores. None of the three is pooled into the analytical corpus.

What this means for the test. Because rejected and withdrawn submissions are present, what topic newcomers write is observed for every paper they submitted, and the gate tests compare accepted with rejected papers of the same topic and year. Neither half of the herding account is tested on survivors only.

B Measurement Validation

Each prediction that separates imitation from continuity is measured with four instruments: topics, contribution labels, entry status, and distances in one embedding space. This appendix reports how far each can be trusted.

Topics. Title and abstract ("title [SEP] abstract") are embedded with SPECTER2 (Cohan et al., 2020; Singh et al., 2023) (base model with the proximity adapter), reduced by unit-length PCA to 128 dimensions (93% of the variance), and clustered by spherical k-means with $k = 100$ (best of 3 seeds $\times$ 20 initialisations). A broad level (20 topics, Ward) and a fine level (198 topics) are supplementary. The topic layer reads no acceptance, decision or review field; an output validator checks this. Topics hold 127–1,121 papers; 91 topics (94% of papers) have at least 30 papers in at least three years. On 2024–2026 papers, normalised mutual information (NMI) with ICLR's primary areas is 0.302, against 0.272 for a TF-IDF/LSA word-count baseline, and synonym overlap is 0.67 against 0.46. In a blind name check, the correct topic was picked for 80% of items at this level and for 61% at the fine level (chance 10%). The adjusted Rand index (ARI) across seeds is 0.66 (0.63–0.69); the median Jaccard overlap is 0.66, with 26 topics below 0.5. Table 4 gives the sensitivity to k: $k = 100$ was fixed before the sweep (a design choice, not tuned), and adjacent solutions nest only partly (76–82% of a finer topic's papers fall in one coarser topic). Table 5 shows that the registered estimates keep their sign at $k = 75$ and $k = 150$, except the inconclusive H2a at $k = 150$.

k	Track.	NMI	ARI	Syn.	Coh.	NMI_{100}
50	100%	0.294	0.77	0.77	0.26	0.72
75	97%	0.302	0.74	0.72	0.27	0.80
100	**94%**	**0.302**	**0.68**	**0.67**	**0.29**	1
150	76%	0.301	0.62	0.67	0.31	0.78
200	53%	0.302	0.57	0.64	0.33	0.75

Table 4: Topic quality by k (sensitivity sweep). Track.: papers in trackable topics; NMI with ICLR primary areas; ARI across seeds in the sweep (the main map's own seed ARI is 0.66); Syn.: synonym overlap; Coh.: keyword coherence; NMI_{100}: NMI with the $k = 100$ map. Outcome-free (E).

Test (unit)	$k = 75$	$k = 100$	$k = 150$
H1d (pp, 0→100% FM)	−8.70	−7.98	−7.47
H2b (pp)	+3.06	+3.71	+3.87
H2c (pp)	−1.51	−1.70	−1.81
H2a (pp)	−1.39	−0.73	+0.37
H3a (pp per log unit)	−1.54	−1.68	−1.33

Table 5: Registered discovery estimates at three values of k (counted variants). Intervals for $k = 100$ are in Table 10. H3a is inconclusive at every k.

Contribution labels. Each paper receives one primary contribution type from a fixed LLM codebook (prompt v1): method (including new

[3] https://github.com/MlouisBE/iclr-rebuttal-analysis

[4] Described in arXiv:2510.13201.

architectures, type "model"), theory, empirical, dataset/benchmark, evaluation, other. A measurement paper is a dataset/benchmark or evaluation paper. An FM paper uses or adapts a large pre-trained model (an LLM, CLIP or a pre-trained diffusion model), coded with the same codebook. Labels were produced by Claude Haiku subagents with the prompt unchanged, in batches of 100; this replaced a planned external run and is a logged deviation. The registered quality gate required at least 93% agreement with a 300-paper pilot and a second-pass $\kappa \geq 0.80$; the run achieved 97.7% and κ 0.93. Against a reference set of 2,300 papers labelled by LLM annotators, main-class agreement is 95.8% (pilot 300: 97.7%; 200 double-labelled training papers: 95.5%; 1,800 single-labelled: 95.6%). A blind second pass on 2,174 discovery papers gives κ 0.93 overall, 0.933 for method (97.5% agreement) and 0.949 for measurement (99.3%). Relative to the LLM reference on its random stratum, false-positive and false-negative rates are 0.084 and 0.011 for method, and 0.005 and 0.070 for measurement; the S8 variant rescales the entry estimates by them, which can only enlarge them (+4.09 and -1.84 pp). On 555 held-out papers, agreement is 96.9% for the main class (κ 0.931), 98.0% for method (0.949), 99.1% for measurement (0.923) and 98.2% for FM (0.960). These figures show consistency between LLM labellings more than human-level validity. Over all discovery papers, the class mix is method 75.1%, empirical 6.9%, theory 6.9%, dataset/benchmark 5.2%, other 3.5% and evaluation 2.4%; the measurement share rose from 2.6% (2017) to 11.1% (2026) (E).

Keywords. Author keywords are canonicalised to 40,930 expressions. The 912 keywords with at least 20 papers were typed by two LLM annotators (κ 0.93; 96.6% agreement; 31 adjudicated). Keyword states use a 759-keyword vocabulary. Held-out papers carry raw keywords only; a lookup learned on discovery papers alone maps 45,778 forms (5,738 of them non-identity), with cross-validated recall and precision of 98.6% and 99.96% on the 759-keyword vocabulary, 99.0% and 99.98% on the 68 frequent keywords, and 97.2% and 100% for "federated learning".

Entry status. A paper is an incumbent paper if at least one author identifier has an earlier ICLR submission, with any outcome, in the paper's topic, strictly before the paper's year; otherwise it is a newcomer paper. Exact identifiers are used, never names. ICLR author identifiers were 98–99.6% e-mail-based in 2017–2020, 85% profile-based in 2021 and fully profile-based in 2025–2026, and the two kinds are not linked. Earlier histories are therefore hidden after 2020: incumbency on the all-years map falls from 18% (2020) to 0.5% (2021) and recovers to 18% (2022) and 33% (2026). This turns some true incumbents into newcomers and pulls entry contrasts towards zero. The counted R2 variant drops only 2021–2022 and leaves the entry results unchanged: +3.72 [+2.61, +4.83] and -1.74 [-2.53, -0.95] pp (n 37,312). Earlier work that sits in the held-out split is invisible to discovery histories, which also pulls towards zero. Held-out features were built on pooled histories (held-out incumbency 30.9%). Discovery incumbency is 26.5% under discovery-only histories; under pooled histories it would be 30.0%, with 1,515 discovery papers switching from newcomer to incumbent. Experience at other venues is not captured.

Probable earlier versions. 170 discovery papers are flagged as probable resubmissions: 167 have an earlier paper at centred SPECTER2 cosine $\geq \tau = 0.90$, and 3 an earlier paper with an identical normalised title. The threshold was calibrated by blind LLM judges (two Haiku runs and one Sonnet run), whose yes-rates were 10%, 48%, 90% and 100% across four cosine bins. The flags define the R1 and S4 variants and the exclusions in both distance measures.

Distances. Both distances use SPECTER2 vectors centred with a frozen discovery mean and L2-normalised. Closeness to the team's own record ($F_{\max}$) is, for each experienced author, the cosine between the paper's vector and the normalised mean of that author's earlier ICLR submissions since 2021, excluding the paper's probable earlier version and any paper at cosine ≥ 0.90; $F_{\max}$ is the maximum over experienced authors. The measure follows Bao and Li (2025). Distance from earlier ICLR work is $1-$ the mean cosine to the 10 nearest submissions from strictly earlier years, as a percentile within topic $\times$ year (Appendix E). The space is the same; the reference points differ.

A dropped measure. A registered measure of buzzword use (a fast-growing keyword used atypically) failed its blind validation: LLM judges rated 42% (95% CI 33–52%) of flagged uses peripheral, against 15% of matched non-flagged uses, and the

registered gate required at least 50%. Its two hypotheses (H2d, H3c) were dropped before any estimate; exploratory rows that use it are reported and not interpreted.

Held-out topic maps. For each of nine alternative maps ($k = 50$, 75, 150 and 200, the seed refit, the year-balanced map and the maps fitted on 2017–2020, 2021–2023 and 2024–2026), the exact centres were re-derived from the frozen fits; every re-derivation reproduces all saved discovery labels. On every map the held-out topic distribution differs from discovery by no more than sampling noise, and the share of stable papers is 0.792 in both splits.

What this means for the test. The labels are consistent rather than validated against human judgement, and the identifier switch hides earlier histories. The switch pulls the entry contrasts towards zero, as does label error if it is non-differential, so neither can produce the reversal of the registered herding prediction.

C Hindsight Details

The herding reading starts from vivid waves. Section 4 shows that the number of emerging topics a map finds depends on the years used to fit it. This appendix gives the six maps in full and three further results that depend on the fitting years.

Six maps on the same papers. All maps use $k = 100$, the same 43,480 papers and the same pipeline; only the fitting years differ (Table 6). A topic is emerging if its 2024–2026 share is at least three times its 2017–2020 share, and declining if it is at most a third. Because 69% of papers are from 2024–2026, the all-years map is in effect a present-day map: it fits 2026 papers better than 2017 papers (mean similarity to the assigned centroid 0.65 against 0.62), while the 2017–2020 map does the reverse (0.69 against 0.53). Growth compounds hindsight: the all-years map finds as many emerging topics as the map fitted on 2024–2026 alone (29 against 28). The share of 2024–2026 papers in emerging topics brackets the effect, from 22.3% on the 2017–2020 map to 40.9% on the all-years map (20.9–42.2% across the six maps), rather than pinning it down: the early map cannot represent topics born after 2020, and the maps differ in fitting corpus as well as in window (E).

Map (fitted on)	ARI	Emerging	Declining	Fit 2017/26
All years	1	29 (40.9)	13 (41.7)	0.62/0.65
Seed refit	0.69	28 (40.9)	15 (43.2)	0.62/0.65
Year-balanced	0.48	16 (42.2)	26 (38.6)	0.69/0.61
2024–2026	0.55	28 (31.0)	6 (30.2)	0.56/0.66
2021–2023	0.40	10 (20.9)	17 (31.6)	0.63/0.58
2017–2020	0.34	8 (22.3)	17 (21.9)	0.69/0.53

Table 6: Six $k = 100$ maps that differ only in their fitting years. ARI with the all-years map. Emerging: count (share of 2024–2026 papers in them, %). Declining: count (share of 2017–2020 papers in them, %). Fit: mean similarity of a year's papers to their assigned centroid. Outcome-free (E).

Year	All	Seed	Bal.	24–26	21–23	17–20
2017	0.443	0.470	0.636	0.448	0.556	0.738
2018	0.525	0.529	0.743	0.496	0.655	0.850
2019	0.582	0.589	0.808	0.520	0.745	0.888
2020	0.660	0.665	0.850	0.574	0.833	**0.908**
2021	0.703	0.713	**0.867**	0.619	0.915	0.867
2022	0.725	0.731	0.831	0.629	**0.921**	0.804
2023	0.797	0.795	0.847	0.697	0.911	0.785
2024	0.897	0.893	0.820	0.842	0.839	0.792
2025	**0.910**	**0.896**	0.731	**0.937**	0.739	0.731
2026	0.840	0.831	0.634	0.930	0.640	0.681

Table 7: Effective number of topics divided by k, by year and map (maximum in bold). Each map is most diverse in its own fitting years. Outcome-free (E); not a finding.

Diversity is not a finding. The effective number of topics per year (as a share of k) peaks inside each map's own fitting years (Table 7): in 2020 on the 2017–2020 map, in 2022 on the 2021–2023 map, in 2025 on the 2024–2026 and all-years maps, and in 2021 on the year-balanced map. The all-years map's rise in diversity to 2025 is therefore a property of where the map was fitted, not a finding about the field.

Topic growth looks predictable only in hindsight. An out-of-time forecast of two-year topic growth (test origins 2021–2024; post hoc, outcome-free) shows the same asymmetry (Table 8). On the all-years map, small topics grow: topic size "predicts" growth with R^2 0.28. On the 2017–2020 map it does not (R^2 −0.03). The in-sample chance of top-fifth growth by size tercile is 40%, 21% and 2% (small, middle, large) on the all-years map, against 22%, 20% and 20% on the early map. Caveats: the early map cannot represent topics born after 2020, the maps differ in fitting corpus and eligibility timing, so the gap cannot be attributed to hindsight alone, and there is no paired test. Genuine fore-

Predictor: metric	All years	2017–2020
Size: R^2	0.281	−0.033
Size: Spearman	0.595	−0.043
Size: precision, top fifth	0.563	0.250
Size: R^2 without LLM/VLM areas	0.30	−0.09
Momentum: R^2	0.056	0.032
Momentum: Spearman	0.377	0.198
Momentum: precision, top fifth	0.420	0.395
Topic-years	446	522

Table 8: Out-of-time forecasts of two-year topic growth, test origins 2021–2024. Intervals for size: R^2 [0.217, 0.335] against [−0.082, −0.002]; Spearman [0.510, 0.672] against [−0.151, 0.063]. Chance precision is 0.20. Post hoc, outcome-free (E).

Variant	H1a (−)	H1b (+)	H1c (+)	n
All-years (main)	−0.14	+0.50	−0.31	43,096
$k = 75$	−0.10	+0.93	−0.74	43,096
$k = 150$	−0.35	+0.69	−0.13	43,096
Year-balanced^h	−0.08	+0.66	−0.75	43,096
Early map, all papers	−2.30	+2.16	−1.06	43,096
Early map, 2021–2026^h	−3.48	+3.96	−1.03	39,008
Stable papers	+0.53	+0.28	−0.37	34,144
Coherent topics	−0.20	+0.50	−0.26	42,161
Certain labels	−0.66	+0.92	−0.56	26,369
Without 2026	+0.48	−0.53	−0.16	27,685

Table 9: Topic crowding and contribution mix: every counted variant of the registered H1a (method share), H1b (measurement share) and H1c (empirical share), pp per log unit of prior topic papers; registered sign in parentheses; ^{h} hindsight variant; early map: fitted on 2017–2020. Predicted sign: 7/9, 8/9 and 0/9 variants. Outcome-free labels.

casting is weak: on the early map, momentum's top-fifth picks are right for only 16 of 56 (29%) for origins 2021–2023.

The lifecycle answer depends on the map. The registered tests of whether topics "age" from method to measurement work (H1a–c) show nothing on the all-years map: method −0.14 [−1.78, +1.49] and measurement +0.50 [−0.76, +1.75] pp per log unit of prior papers (C). On the 2017–2020 map applied to 2021–2026 papers, the method and measurement estimates point the predicted way: −3.48 [−6.21, −0.76] and +3.96 [+1.35, +6.57] (counted variants, read as a lifecycle only in exploratory and unadjusted terms). The empirical share (H1c) has the wrong sign on every map. Table 9 lists every counted variant. By the registered rule all three are not supported, because the main map shows nothing; the lesson is that the answer depends on the map.

What this means for the test. A herd reading starts from the waves, and the emergence a field history shows is partly a property of when its map was fitted. For this reason every registered time-trend claim had to hold on the year-balanced map and on the 2017–2020 map applied to 2021–2026 papers. H1d, the package result, passes both; the lifecycle claims do not.

D Every Registered and Declared Test

The negative half of our thesis rests on registered tests and the positive half on post hoc tests with declared held-out tests. This appendix lists every such test, including those that failed, so that the denominators are visible (Table 10).

Protocol. Hypotheses, models, the support rule and the robustness variants were registered on 2026-10-13, before any full-corpus label existed (only the 2,300 validation papers had labels, and no hypothesis had been tested on them). Earlier exploratory work on keyword popularity and title style in the same corpus came before registration and is disclosed. A hypothesis is *supported* only if its main estimate has the predicted sign, its Holm-adjusted p is below 0.05 within its family (Holm, 1979), at least 75% of its counted variants have the predicted sign, and both hindsight variants (year-balanced map; 2017–2020 map on 2021–2026 papers) have the predicted sign. Otherwise the verdict is "significant in the opposite direction" (Holm $p < 0.05$ with the wrong sign), "equivalent to null" (90% CI inside the smallest effect of interest) or "inconclusive". Discovery Holm families were {H1a–e}, {H2a–c} and {H3a–b}. Linear probability models use topic and year (or topic × year) fixed effects with standard errors clustered by topic (100 clusters). Outcomes were readable only through a sealed loader, which ran once inside the frozen run (18 seconds). Nine adversarial review rounds preceded the freeze; two blinding incidents are disclosed (a reviewer saw one exploratory estimate on the failed buzzword measure, and the analyst once read the value sets of outcome columns).

Held-out tests. Held-out tests were registered on 2026-10-18 and run once (16:39–16:42 UTC). Every test is one-sided at $\alpha = 0.05$: $p_1 = p/2$ if the estimate has the predicted sign, otherwise $1 - p/2$, where p is the two-sided p of the discovery method; the exception is premia b, whose p_1 is the intersection-union p for $0 < b < 1$ (0.003). Holm

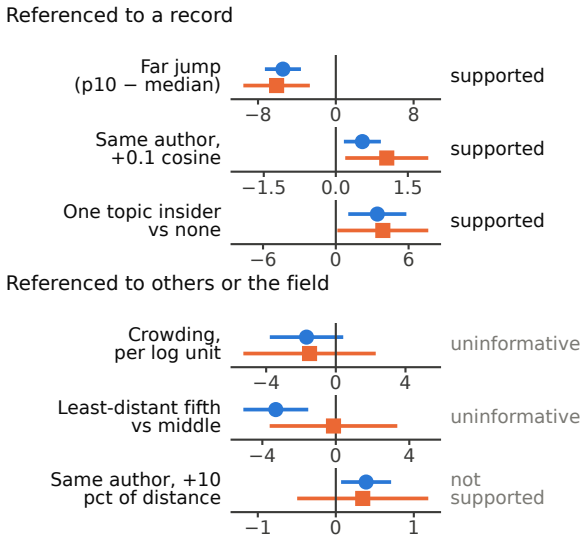


Figure 5: Gate tests by reference point: discovery (blue circles), held-out (orange squares), 95% intervals, each row on its own axis, held-out verdict at right. 'Others or the field' groups crowding (others now) with distance from earlier ICLR work (the field's record); 'Same author, +10 pct of distance' is the within-author slope for that distance. Compare replication status, not size. All rows except crowding (registered H3a) were post hoc in discovery; held-out tests are one-sided, and the same-author and insider tests are secondary and unadjusted.

families were {H1d}, {H2b′, H2c′}, {H3a}, {H4-FL} and the post hoc primary pair {E1, P1b}. Six post hoc secondary tests (E4, P7, teams H1, A1, premia b, premia D0) are each one-sided at 0.05 and unadjusted, so one false positive among them would not be surprising; their Holm values hold within their own module only (teams H1 about 0.17 across modules, A1 about 0.18). The PH6 wave test was declared on 2026-10-28 and run once. H2b′ and H2c′ reverse the registered herding predictions, which failed in discovery; they were registered after discovery and before any held-out data existed.

Discovery details. H1a–c, H1e: Holm 1.0; predicted sign in 7/9, 8/9, 0/9 and 4/9 counted variants (H1c also fails both hindsight variants). H1d: Holm 0.0014; 9/9 variants (−6.3 to −14.2). H2a: Holm 0.46; 8/10 variants with the predicted sign, but the year-balanced hindsight variant has the wrong sign (+1.05), as does $k = 150$ (+0.37); the two early-map variants are −1.62 (p 0.03) and −1.55 (p 0.045); TOST p 0.10. H2b and H2c: wild cluster restricted bootstrap $p < 1/9{,}999$, Holm < 0.001 with the wrong sign; all 10 counted variants share the observed sign (+2.16 to +3.87; −1.81 to −0.86), as do both hindsight variants (+3.10 and +2.16; −1.41 and −0.86). With log team size, H2c shrinks to −0.54 [−1.11, +0.04] (p 0.07) while H2b stays at +3.92; by era, H2c is concentrated in 2024–2026 (−2.0, against −0.5 n.s. in each earlier era). H3a: Holm 0.20; all 11 counted variants negative (−2.53 to −1.31; logit average marginal effect −1.64); supplementary variants are negative and some nominally significant (team size −2.04, fine topics −2.42, withdrawals after review counted as rejections −2.12, newcomer control −2.99), but they are unadjusted and outside the verdict; TOST p 0.75. H3b: Holm 0.53; 10/11 variants negative; TOST p 0.14. No non-supported hypothesis is equivalent to null.

Held-out details. H1d: 9/9 variants negative (−5.4 to −15.2); the weakest, the prospective early-map variant, is −5.4 (two-sided p 0.24). H2b′ and H2c′: 10/10 variants, both hindsight variants pass; two-way clustering agrees (Holm p_1 0.0010 and 0.0009); the stress sensitivity gives Holm p_1 0.0075 for both; no variant is significant against the prediction. With log team size, H2c′ shrinks to −0.98 (two-sided p 0.10) while H2b′ stays at +3.64, as in discovery. H3a: 9/11 variants negative; the hindsight requirement fails (early map on 2021–2026 papers +1.44); TOST p 0.61. H4-FL: yearly held-out gaps of −20.0, −16.9, +6.4, −3.6, −11.3 and −0.2 pp (2021–2026, on 13–36 papers a year); the discovery pattern of about −11 to −15 pp in every year since 2023 does not appear.

Reading rules. Four supported tests were underpowered at the discovery effect: H1d (MDE80 12.0 against −7.98), H2c′ (2.4 against −1.70), teams H1 (≈4.5 against +3.42) and A1 (≈4.3 against +2.67). They confirm direction, not size; among underpowered tests, the significant ones tend to overstate the effect. A non-supported registered test whose estimate is below its MDE80 (H3a, H4-FL, P1b) is uninformative, not evidence of no association. A1's 95% interval includes 0 because a one-sided 5% test corresponds to a 90% interval.

Further notes to Table 1. Row 6 (distance from earlier ICLR work): outcome-free, unregistered and added during writing; among method papers the gap is +5.24 [+4.16, +6.31]; "own work excluded" means that every earlier paper sharing an author with the paper is removed from its reference set (Appendix E). Row 7: the FM gaps compare first-time teams with returning topic newcomers; no gap is detectable at 0–5% (−0.95) or ≥50%

Test (predicted sign; held-out n)	Discovery [95% CI]	Held-out [95% CI]	p_1	MDE80	Verdict (tier)
Field: registered 2026-10-13; H1d also registered for held-out papers					
H1a method share ∼ crowding (−)	−0.14 [−1.78, +1.49]	not tested			not supported (C)
H1b measurement share ∼ crowding (+)	+0.50 [−0.76, +1.75]	not tested			not supported (C)
H1c empirical share ∼ crowding (+)	−0.31 [−1.29, +0.67]	not tested			not supported (C)
H1d new-architecture share ∼ prior FM adoption, 0→100% (−; 10,978)	−7.98 [−12.14, −3.82]	−11.0 [−18.1, −3.9]	0.0013	12.0	supported (A)
H1e empirical or evaluation share ∼ prior FM adoption (+)	−0.63 [−3.64, +2.38]	not tested			not supported (C)
Entry: registered 2026-10-13; reversed H2b′ and H2c′ registered for held-out papers after discovery					
H2a incumbent share ∼ hot topic (−)	−0.73 [−2.68, +1.22]	not registered[u]		3.6[f]	inconclusive (C)
H2b method paper ∼ newcomer paper (−); reversed H2b′ (+; 10,718)	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	0.0007	2.8 (1.7[f])	failed; reversal supported (A)
H2c measurement paper ∼ newcomer paper (+); reversed H2c′ (−; 10,718)	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	0.0005	2.4 (1.1[f])	failed; reversal supported (A)
Gate: registered 2026-10-13 (H3a also for held-out papers); H4-FL registered 2026-10-17, before any held-out data					
H3a acceptance ∼ crowding, per log unit (−; 8,185)	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	0.21	4.8	inconclusive; uninformative (C)
H3b acceptance ∼ crowding × method paper (−)	−0.36 [−1.51, +0.79]	not registered[u]		2.4[f]	inconclusive (C)
H4-FL federated-learning keyword, level (−; 134 FL papers)	−11.3 [−17.9, −4.8]	−4.9 [−15.4, +5.7]	0.14	14.5	not supported; uninformative (C; discovery D)
H4-FL trend, pp per year (−)	−3.0 [−6.4, +0.5]	+3.2 [−3.3, +9.6]	0.88	n.c.	not supported, opposite sign (C)
Gate: post hoc in discovery (declared before run); held-out tests declared 2026-10-18 (PH6: 2026-10-28)					
E1 far jump: $F_{\max}$ p10 − p50 (−; 5,344)	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	0.0025	4.9 (0.89)	supported, primary (B)
P1b field distance: least-distant fifth − middle fifth (−; 8,010)	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	0.47	5.4 (0.47)	not supported; uninformative (C; discovery D)
E4 within-author closeness, per +0.1 cosine (+; 4,311)	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	0.006	n.c.	supported, secondary (B)
P7 within-author field distance, per 10 percentile points (+; 4,810)	+0.39 [+0.09, +0.69]	+0.34 [−0.47, +1.16]	0.22	≈1.0–1.1[d]	not supported (C)
Teams H1: one topic insider vs none (+; 6,243)	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	0.017	≈4.5[d]	supported, secondary (B)
A1 Emerging − Mainstream keyword (+; 5,741)	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	0.031[n]	≈4.3[d]	supported, weak: CI includes 0 (B weak)
Premia b: 2024–2026 topic edge on 2018–2022 edge ($0 < b < 1$; 75 units)	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	0.003[i]	n.c.	supported, secondary (B)
Premia D0: EB-shrunk edges predict better than zero, $\times 10^{-3}$ (+)	none (new test)	1.71 [0.42, 3.01]	0.005	n.c.	supported, secondary (B)
PH6 waves: gap after − up to a keyword's share peak (−; 17 waves)	−5.7 [−8.5, −2.9][t]	−5.2 [−11.1, +0.7]	0.043	7.4	supported, only just: CI includes 0 (B weak)

Table 10: Every registered and declared test. Estimates in pp within topic and year (H1d: change in the new-architecture share for prior-year FM adoption from 0 to 100%; keyword and wave rows: against all other papers of the same year). Discovery intervals ±1.96 SE; the held-out H1d interval uses $t(G-1)$ quantiles. p_1: held-out one-sided p; Holm-adjusted, it is unchanged except for H2b′ and H2c′ (0.0010 each) and E1 (0.0050). MDE80: held-out minimum detectable effect at 80% power, registered unless marked, with power at the discovery effect for E1 and P1b in parentheses; [f] discovery MDE80 from the frozen dry run; [d] derived post hoc (2.49×SE), with no reading rule; n.c. not computed. [u] unregistered held-out row (Table 12). [n] null-frame p_1 0.020; two-sided p 0.063. [i] intersection-union p for $0 < b < 1$. [t] not a test. Tiers as in Table 1; reading rules in the text.

Split (waves)	D	95% interval	p_1	MDE80
Held-out, declared test (17)	−5.2	[−11.1, +0.7]	0.043	7.4
Held-out, without six inspected (11)	−1.7	[−9.7, +6.3]	0.33	10.1
Held-out, without FL (16)	−6.2	[−12.5, +0.1]	0.033	7.9
Discovery, not a test (17)	−5.7	[−8.5, −2.9]	<0.001	3.6
Discovery, without six inspected (11)	−3.7	[−7.5, +0.2]	0.034	4.8

Table 11: PH6: weighted mean over waves of a keyword's acceptance gap after minus up to its share peak, pp; 2,000 within-year permutations. Held-out papers before and after peaks: 454 and 1,022.

(−1.96) prior adoption, or between returning topic newcomers and incumbents in any bin (−2.58 to +2.87); the keyword contrast compares first-time with returning teams and has no team-size control. Row 10: with keywords as units the held-out slope is +0.16 [−0.34, +0.66] (unregistered); empirical-Bayes-shrunk edges predict better than zero (D0 1.71×10^{-3}, B). Row 16: estimated at fixed team size and number of experienced authors; underpowered at the discovery effect; its Holm value holds within its module only (≈0.17 across modules); an early-map insider gives +3.0 and a keyword-defined insider +0.0.

PH6 variants. The wave test is post hoc: the question came from viewing the discovery per-year gaps of six keywords, and its wave set is retrospective by construction. Table 11 shows that it leans on those six keywords: without them the held-out estimate is −1.7 (p_1 0.33). Vision transformer has the largest drop in both splits, and federated learning itself does not decline in the held-out papers (+3.5 pp after against up to its peak; discovery −5.9), consistent with the failed H4-FL trend.

Unregistered held-out rows. The held-out run also produced 17 rows that were not registered (Table 12). They are two-sided and unadjusted, and they are reported, not interpreted. One quoting rule follows from them, in either direction: a discovery reading that rests on an unregistered contrast is quoted only with its held-out and pooled rows (P2 and E2 in Appendix F).

Pooled re-run (confirms nothing). After the held-out run, every analysis was re-run on the pooled corpus (54,579 papers). It confirms nothing: about 80% of each pooled sample is discovery papers, so agreement with discovery is expected. Pooling also changes discovery features: hot status changes for 8.1% of discovery papers, discovery incumbency rises from 26.5% to 30.0%, probable earlier versions rise from 170 to 224, and crowding rises by about 0.22 log units. Table 13 gives the main rows. Every registered estimate keeps its discovery sign and descriptive verdict except H3b, which moves to zero (TOST p 0.034, labelled "equivalent to null" only descriptively); H1a −0.59, H1b +0.81, H1c −0.24, H1e −0.6 and H2a −0.85 remain not supported or inconclusive. Four topics pass Benjamini–Hochberg correction in the pooled area analysis (clinical ML −16.2, federated learning −12.8, graph representation learning −9.7, robot learning +12.1); no keyword does. The pooled P1b rests mainly on the discovery papers.

What this means for the test. Read by reference point, the denominators line up with the thesis. The package convergence (H1d) replicated but does not discriminate between the routes. The registered imitation-side tests either failed in the opposite direction (H2b, H2c) or were inconclusive (H2a; H3a, which predicted a crowding penalty and whose intervals exclude premiums above +0.32 and +2.19 pp per log unit: no detectable premium). Among the post hoc gate patterns taken to held-out papers, those referenced to a record held (E1, E4, teams H1) and topic edges regressed (premia b, D0); the timing tests held only weakly (A1, PH6); those referenced to the field or to one vivid case did not (P1b, P7, H4-FL).

E Distance from Earlier ICLR Work

The follow-on reading of herding predicts that topic newcomers write work that sits close to what the topic has already produced. This appendix gives the full comparison behind Section 5. It was added during writing, reads no outcome, is unregistered, and is descriptive (E).

Definition. Vectors are SPECTER2 embeddings of title and abstract, centred with the frozen discovery mean and L2-normalised. A paper's distance from earlier ICLR work is 1− the mean cosine to its 10 nearest discovery submissions from strictly earlier years (any topic), with the paper's own chain of probable earlier versions removed from the reference set. It is then ranked within the paper's topic × year cell over all submissions in the cell and reported in percentile points. The measure reim-

Row (sign of interest)	Discovery [95% CI]	Held-out [95% CI]	p
H2a incumbent share ∼ hot topic (−)	−0.73 [−2.68, +1.22]	−2.55 [−5.59, +0.49]	0.099
H3b acceptance ∼ crowding × method (−)	−0.36 [−1.51, +0.79]	+1.34 [−1.26, +3.93]	0.31
H4-FL recent level, 2024–2026	−13.5 (p 0.007)	−3.6 [−17.2, +9.9]	0.52
H4-FL level adjusted for class, FM and team size	−10.5	−3.4	0.42
E2 near: $F_{\max}$ p90 − p50	+1.75 [−0.17, +3.66]	+2.34 [−1.01, +5.70]	0.17
E3 far jump, hot minus other topics	−0.35 [−3.13, +2.42]	−3.09 [−9.63, +3.45]	0.35
P1a field distance, most-distant fifth − middle fifth	+0.44 [−1.41, +2.28]	+2.54 [−1.40, +6.48]	0.20
P2 any independent same-year twin	+0.68 [−0.84, +2.20]	+3.69 [+0.90, +6.48]	0.010
P3 twin association, high minus low field distance	+0.18 [−3.46, +3.83]	+4.84 [−2.27, +11.95]	0.19
P4 twin association, hot minus other topics	−1.49 [−4.51, +1.52]	+1.47 [−3.65, +6.59]	0.63
P5 (Q5 − Q3) field distance, newcomer minus incumbent papers	+1.32 [−2.63, +5.28]	−9.75 [−17.38, −2.12]	0.030
P6 (Q5 − Q3) field distance, mature minus young topics	+4.69 [+0.21, +9.18]	+10.74 [+1.95, +19.53]	0.030
Teams H2 concavity: (1 vs 0) − (2+ vs 1) insiders	+2.95 [−1.41, +7.30]	+4.47 [−2.24, +11.18]	0.19
Premia b, all-years-map topics (62 units)	0.35 [0.17, 0.54]	0.51 [0.23, 0.79]	0.0004^{i}
Premia b, keywords (35 units)	0.53 [0.10, 0.95]	+0.16 [−0.34, +0.66]	0.26^{i}
Premia b, areas (20 units)	0.20 [−0.49, 0.89]	0.71 [0.09, 1.33]	0.17^{i}
Premia D = MSE(raw) − MSE(EB), $\times 10^{-3}$ (descriptive)	none	0.37 [−1.31, +2.06]	0.33^{o}

Table 12: Unregistered held-out rows: reported, not interpreted. Two-sided, unadjusted p; ^{i} intersection-union p; ^{o} one-sided. Estimates in pp unless noted. P5 and P6 change sign or size between splits and are not interpreted; they are listed so that the count of 17 rows is complete.

Analysis	Discovery	Pooled
H1d (0→100% FM)	−8.0	−8.9
H2b method ∼ newcomer	+3.71	+3.46
H2c measurement ∼ newcomer	−1.70	−1.65
H3a per log unit	−1.68	−1.72
H3b	−0.36	−0.03
FM papers, acceptance	+4.62	+5.61
Newcomer papers, acceptance	−10.2	−10.2
New arch. × prior FM	+8.3	+9.4
Team size, per doubling	+8.8	+8.8
Newcomer, author FE	−3.1	−3.1
FL keyword, level	−11.3	−9.9
FL keyword, trend per year	−3.0	−1.9
E1 far jump	−5.44	−5.53
E2 near	+1.75	+2.10
P1b least-distant fifth	−3.27	−2.97
P2 any twin	+0.68	+1.86
E4 within-author closeness	+0.55	+0.78
P7 within-author field distance	+0.39	+0.43
Teams H1 one insider	+3.42	+2.90
A1 Emerging − Mainstream	+2.67	+3.10
Premia b	0.38	0.45

Table 13: Pooled re-run (exploratory; confirms nothing). pp unless noted. Pooled Holm or q values: H1d 3×10^{-4}; H3a 0.098; FL level q 0.14 (trend p 0.18); E2 0.036; P2 0.063. The discovery outputs remain the registered record.

plements the one used by the gate's field-distance tests (P1b, P7) at $K = 10$, without importing that module. The population is the entry population (2019–2026 papers with an author identifier; 42,325 papers, 786 cells, 100 topics). The contrast is identified by the 575 cells that contain both newcomer and incumbent papers (39,500 papers). Estimates come from linear models with topic × year fixed effects and standard errors clustered by topic (CR1, t with $G - 1$ degrees of freedom). In the own-work-excluded variant, every earlier paper that shares an author key with the paper is removed from its reference set; author keys are used only as join keys and never written out. The script and its outputs are committed with the study's code.

Main result. Within topic and year, newcomer papers sit +5.90 percentile points [+4.95, +6.85] farther from earlier ICLR work than incumbent papers (mean percentile 51.4 against 46.1; E). On the raw scale the gap is +1.35 [+1.16, +1.55] hundredths of cosine distance, 0.22 of a within-cell standard deviation (0.062): modest but precise. Table 14 gives every specification.

Own work excluded. With every earlier paper by any of the paper's own authors removed from the reference set, newcomer papers still sit +4.33 percentile points [+3.44, +5.22] farther from earlier work than incumbent papers (E); among method papers the gap is +3.71 [+2.70, +4.72], and with team size +4.01 [+3.13, +4.89]. About a quarter of the main gap is therefore the incumbents' closeness to their own earlier work, and the rest is distance from the field's earlier work. The gap remains when the authors' own earlier papers are excluded.

Contribution type, team size and era. The gap holds within method papers (+5.24 [+4.16, +6.31]) and within method papers other than new architectures (+5.00 [+3.86, +6.14]), so it is not the

Specification	Gap [95% CI]	N
Main	+5.90 [+4.95, +6.85]	42,325
Method papers	+5.24 [+4.16, +6.31]	31,764
Method papers, new architectures excluded	+5.00 [+3.86, +6.14]	27,910
Log_2 team size	+5.59 [+4.64, +6.54]	42,325
Class, FM and team-size controls	+5.76 [+4.82, +6.70]	42,325
Method papers, team size	+5.19 [+4.12, +6.26]	31,764
2019–2022	+4.95 [+2.32, +7.58]	8,327
2023–2026	+6.01 [+5.03, +6.98]	33,998
2019–2022, team size	+4.68 [+2.15, +7.20]	8,327
2023–2026, team size	+5.69 [+4.71, +6.67]	33,998
Later minus earlier era	+1.06 [−1.64, +3.75]	42,325
Own work excluded	**+4.33 [+3.44, +5.22]**	42,325
Own work excluded, team size	+4.01 [+3.13, +4.89]	42,325
Own work excluded, method papers	+3.71 [+2.70, +4.72]	31,764
Raw distance ×100	+1.35 [+1.16, +1.55]	42,325
Raw ×100, team size	+1.29 [+1.09, +1.49]	42,325
Raw ×100, method papers	+1.19 [+0.97, +1.42]	31,764
Without 2021	+5.87 [+4.92, +6.83]	39,965
Incumbency ignoring own earlier versions	+5.84 [+4.88, +6.79]	42,325
Papers without an earlier version	+5.81 [+4.84, +6.77]	42,155
Plain definition	+6.63 [+5.78, +7.48]	42,325

Table 14: Newcomer minus incumbent papers, distance from earlier ICLR work in percentile points within topic × year (raw rows: hundredths of cosine distance). Plain definition: uncentred vectors, no earlier-version exclusion (correlation with the main measure 0.74; 0.77 for percentiles). Outcome-free, unregistered (E).

method-type reversal restated. Team size, class and FM controls barely move it (+5.59; +5.76). It is no smaller in 2023–2026 (+6.01 [+5.03, +6.98]) than in 2019–2022 (+4.95 [+2.32, +7.58]); the difference is +1.06 [−1.64, +3.75] (p 0.44), even though the FM share rose from 10.6% (2023) to 54.4% (2026) within the later era.

By year. The gap is positive in every year from 2020 to 2026, and between +5.0 and +6.4 in every such year except 2021 (Table 15). In 2019 it is +1.26 [−5.34, +7.86] (p 0.70), on only 35 cells with both kinds of paper. The 2021 estimate rests on 11 incumbent papers, a consequence of the identifier switch, and is not interpreted. Raw distance falls every year, from 0.473 (2019) to 0.369 (2026), because a growing field leaves closer earlier neighbours; this is why the comparison ranks papers within topic × year.

Year	Gap [95% CI]	Cells	New.	Dist.
2019	+1.26 [−5.34, +7.86]	35	86.3	0.473
2020	+5.74 [+2.01, +9.48]	66	81.9	0.443
2021	not interpreted[a]	11	99.5	0.426
2022	+5.14 [+2.03, +8.24]	76	82.1	0.410
2023	+6.22 [+3.93, +8.50]	91	73.5	0.401
2024	+6.28 [+4.49, +8.07]	96	71.2	0.395
2025	+5.04 [+3.50, +6.57]	100	70.1	0.386
2026	+6.40 [+5.13, +7.66]	100	67.2	0.369

Table 15: Gap by year (percentile points). Cells: topic cells with both newcomer and incumbent papers. New.: newcomer share of papers (%). Dist.: mean raw distance from earlier ICLR work. [a] +22.49 [+10.57, +34.40] on 11 incumbent papers. Outcome-free (E).

Caveats. (1) The analysis was added during writing; it is unregistered, outcome-free and descriptive, and it says nothing about acceptance. (2) Incumbents' own earlier papers can be among their nearest earlier neighbours; the own-work-excluded variant removes them and leaves +4.33. (3) Distance in SPECTER2 space is not originality, and vocabulary from other fields may look distant. (4) Papers are ranked within cells of the all-years map only; the comparison was not re-run on 2017–2020-map cells. (5) The identifier switch turns some incumbents into newcomers after 2020, which pulls the gap towards zero. (6) The measure was recomputed rather than read from the gate module's stored values; a value-by-value comparison with them is not part of the committed output.

What this means for the test. The follow-on prediction expects newcomer papers to sit closer to earlier work than incumbent papers. They sit farther, also when the authors' own earlier papers are excluded. The quarter of the gap that is incumbents' closeness to their own earlier work is itself continuity among producers. A method paper is a contribution type, not a measure of novelty, and this distance is not one either; but neither measure shows newcomers following.

F Other Gate Modules

Section 6 asks whether acceptance goes with following the crowd or with a record. The analyses below are discovery only (D) unless stated. Each was registered as exploratory or declared before it was run; all are associations within topic and year, none is advice, and none changes the reading in Section 6. The planned exploratory analyses are labelled "planned" to separate them from the post hoc far-jump tests E1–E4.

The newcomer gap is mostly about who writes. Newcomer papers are accepted less often within topic × year: −10.2 pp [−11.8, −8.7] with class and FM controls (−10.5 in years with consistent identifiers; planned E2; D). Among 31,549 decided 2019–2026 papers, newcomer papers were accepted at 37.2% and incumbent papers at 47.3%. With 160,222 paper-author rows (81,013 authors, 15,579 of whom wrote both kinds of paper), the gap on the same 28,370 papers is −5.4 [−7.1, −3.7] without and −3.1 [−4.5, −1.6] with author fixed effects; the same author entering a topic new to them is −1.4 pp [−2.7, −0.0] (p 0.049; D). First-author and last-author fixed effects give −2.2 [−4.8, +0.3] and −2.9 [−4.6, −1.1]. ICLR is double-blind, and the residual need not reflect how reviewers treat the person: less topic familiarity can make the paper weaker.

What goes with acceptance among newcomer papers. Against newcomer method papers, newcomer theory papers are +10.2 [+7.2, +13.2] and newcomer FM papers +4.1 [+2.0, +6.2]; dataset/benchmark +2.4, evaluation −0.5 and empirical −3.1 are not significant, and other papers are −9.1. Against teams of 3–4 authors, teams of 1–2 are −8.5, 5–6 +3.7 and 7 or more +11.1. Newcomer papers in more crowded topics are −2.8 per log unit (Holm 0.02); in hot topics +1.2 (n.s.). The newcomer gap is present in every subgroup, from −5.1 (theory) and −6.4 (teams of 7+) to −14.6 (dataset/benchmark) and −11.6 (most crowded topics), and is negative in all 20 broad areas (−3.8 to −20.8), none surviving Holm correction (D). Selection may account for these: newcomer theory papers may come from established researchers in other fields, and large teams bring seniority.

Team size. Acceptance rises with team size: 28.7% (1–2 authors; 4,176 papers), 37.6% (3–4; 12,022), 41.6% (5–6; 8,843) and 49.0% (7+; 7,149). Within topic × year the gradient is +8.8 pp per doubling [+8.0, +9.7] (+7.8 with a newcomer control; D). It is +13.5 for dataset/benchmark and evaluation papers, +9.3 for method, +5.5 for empirical and +3.3 (n.s.) for theory; equal slopes are not rejected (calibrated p 0.13). Areas differ (calibrated p 0.035), but the slope is positive in every area. Team size stands in for compute, data, engineering and seniority.

Topic insiders. Population: 24,601 decided 2023–2026 papers (acceptance 41.1%), of which 16,579 have no topic insider, 3,555 one, 2,053 two and 2,414 three or more. Besides the supported H1 (Table 10), the module's family gives: H2, concavity, +2.95 (Holm 0.55); H3, the insider contrast in hot minus other topics, −0.8 [−4.9, +3.2]; H4, a proven outsider minus an unproven insider, +4.6 (Holm 0.088); H6, last- minus first-author insider, −2.5 (Holm 0.55); H5 is not reported. Two or more insiders against one: +0.5; familiarity without an earlier acceptance (767 papers) gives +0.2 [−3.8, +4.2]. Robustness of H1: the logit is consistent; an early-map insider gives +3.0 (p 0.003); without earlier versions +3.4; a keyword-defined insider +0.0, a null the registration anticipated because the two definitions correlate only 0.18–0.29. A per-head decomposition (one specification outside the family; D) gives +1.47 [+0.95, +1.99] per author new to ICLR, +4.77 [+4.23, +5.31] per experienced author from other topics and +5.34 [+4.65, +6.04] per topic insider; experienced outsider minus novice +3.30 (p 2.8×10^{-16}), insider minus experienced outsider +0.58 (p 0.16). Per head, an insider is worth about as much as an experienced author from another topic, so the insider association is compatible with general experience and visible status as well as with topic continuity.

Far jump: shape and robustness. Among 20,430 decided 2023–2026 papers with at least one experienced author (acceptance 44.1%), adjusted acceptance by fifth of $F_{\max}$, far to near, is 39.7 [38.1, 41.2], 41.8 [40.3, 43.3], 44.4 [43.1, 45.8], 47.2 [45.8, 48.6] and 46.9% [45.2, 48.6] (3,855 to 4,265 papers per fifth; D shape). Acceptance is lower the farther a paper sits from the record, and the closest fifth is no higher than the fourth. E1 holds with a newcomer control (−5.0), with at least three prior authors (−4.1), without earlier versions (−5.5), with withdrawals after review counted as rejections (−4.9), on the early map (−5.2), with closeness to the last author's record (−4.6) or the whole team's (−4.7), and on link-removed text (−5.39). $F_{\max}$ correlates −0.57 with the newcomer flag, so E1 partly overlaps the newcomer gap. E2, near against median closeness, is +1.75 [−0.17, +3.66] (Holm 0.148) in discovery; by the quoting rule it comes with the held-out +2.34 [−1.01, +5.70] (p 0.17, unregistered) and the pooled +2.10 (Holm 0.036), so "no further gain near the record" is not established. E3, the far-jump contrast in hot minus other topics, is −0.35 [−3.13, +2.42]. Within-author closeness (E4) may be ex-

plained half or more by co-author composition. Far means far from the visible ICLR record since 2021; co-author choice and pivots that follow failures are not controlled.

Field distance and twins. The field-distance population is 31,549 decided 2019–2026 papers (acceptance 40.1%). P1b is negative in 11 of 11 variants and Holm-significant in 8 (within class × year −2.0, Holm 0.65; prior-density control −2.4, Holm 0.12; fixed reference −2.5, Holm 0.050); link-removed text gives −3.51. It did not replicate (Table 10). P1a, the most-distant fifth against the middle, is +0.44 [−1.41, +2.28], so maximal distance carries no detectable cost at quintile resolution. A paper has a twin if an independent same-year submission lies at cosine ≥ 0.70; 37.7% of 42,326 papers have one (36,863 independent pairs at 0.70–0.80 and 1,400 at ≥ 0.80, 2018–2026), and the rank correlation between distance and twin count runs from −0.47 (2019) to −0.71 (2026) (E). By the P2 quoting rule, the discovery estimate for having a twin, +0.68 [−0.84, +2.20] (Holm 1.0), comes with the held-out +3.69 [+0.90, +6.48] (p 0.010, unregistered) and the pooled +1.86 [+0.47, +3.24] (Holm 0.063); the held-out estimate does not fit a "no detectable gap" reading. Identical ideas (cosine ≥ 0.80; 692 decided pairs) allow only an underpowered shared-fate test (p 0.15).

Interactions with topic heat. No gate or entry association detectably differs in hot topics. Newcomer × hot: method −0.4, measurement −0.6, acceptance +0.2 pp, all n.s. (planned E5). The insider contrast in hot minus other topics is −0.8 [−4.9, +3.2], the far-jump contrast −0.35 [−3.13, +2.42] (held-out −3.09, unregistered), and the twin association −1.49 [−4.51, +1.52] (held-out +1.47, unregistered). By era, the crowding slope is −0.3, −6.5 and −2.9 pp per log unit (2018–2020, 2021–2023, 2024–2026; all n.s.).

FM premium and new architectures against the tide. FM papers are accepted more often within topic × year: +4.62 [+2.79, +6.44] with class controls (+4.06 without; planned E1; D). Among 24,596 decided 2023–2026 papers (9,908 FM papers), the premium does not detectably change with the topic's prior adoption: +0.19 [−0.96, +1.34] per logit unit (p 0.74). In topics with at least 50% prior adoption, the premium is +2.5 [−1.3, +6.3] (Holm 0.38) on the all-years map and +6.0 [+3.1, +8.8] on the early map. New-architecture papers are accepted relatively more often where prior FM adoption is high: +8.3 [+1.3, +15.4] for 0→100% (p 0.02; planned E8; D). With H1d, one possible reading is scarcity; this is a hypothesis, not a test. A fad premium would be expected to fade with adoption; we detect no such fading, which fits rewarded imitation and efficient choice alike.

Acceptance by contribution class. Against method papers within topic × year (planned E10; D): theory +8.9 [+6.5, +11.3], dataset/benchmark +3.9 [+1.1, +6.7], empirical −2.7 [−5.2, −0.3], evaluation −1.9 (n.s.), other −8.0 (n 32,159).

Withdrawal after review. Withdrawal after review rises with crowding, +1.62 pp per log unit [+0.42, +2.81] (p 0.009; 42,214 papers; planned E3; D); withdrawal before review does not (−0.07, n.s.). Counting these withdrawals as rejections moves H3a to −2.12 (unadjusted; outside the verdict).

Areas and federated learning. Acceptance edges of 68 frequent keywords and 100 topics spread about ±10 pp, but none passes Benjamini–Hochberg correction (minimum q 0.185 for keyword levels), and the declared tests were conservative (permutation rejection 0.5–1.4% at nominal 5%). Federated learning was the starkest discovery case: 475 keyword papers, −11.3 [−17.9, −4.8] (p 0.005, q 0.19); topic-defined, 485 papers, −13.5 [−21.1, −5.9] (q 0.17); adjusted −10.5; trend −3.0 pp per year [−6.4, +0.5]. It was not confirmed on held-out papers (Table 10). Three mechanism probes were declared. Composition and influx: not supported; the share of papers with an author with an earlier accepted ICLR paper rose from 4% (2021) to about 60% (2023–2025), keyword newcomers fell from 96% to 40–58%, team size (4.3–4.9) and the method share (about 85%) were stable, and composition adjustment keeps 76% of the raw trend (adjusted −2.2 pp per year, p 0.20). The field moved on: supported descriptively, with the keyword's share peaking at 3.2% (2023) and falling to 1.1% (2026). Harsher judgement: consistent with the data, not testable here; the decline appears in every author group (none significant), and the registered reading rule returns "mixed".

Keyword timing. The A1 population is 22,302 decided 2019–2026 papers (40,074 keyword rows); the Mainstream acceptance rate is 37.8%. A1 is positive in 7 of 7 variants (+2.3 to +4.1, and the

logit). Cooling minus Mainstream is +0.13 (Holm 0.90), and the age gradient +1.74 (Holm 0.077), not supported. Old rising keywords give +3.9 nominally, so A1 may reflect a low Mainstream baseline as much as keyword age; within a keyword, rising against flat years give +0.9 [−0.8, +2.7]. Age, size and breadth of keywords are confounded. In PH6, relative to each wave's own level, the discovery gap is +9 to +30 pp 2–5 years before the peak (on few papers), near 0 around the peak and −2 to −5 pp 3–5 years after it: the decline starts before the peak.

Topic edges and the keyword winner's curse. The noise-free SD of topic acceptance edges is 6.4 pp [5.0, 7.2] on the early map (6.1 on the all-years map; keywords 4.7; areas 4.4); a single year's rate is half noise at about 58 decided papers. The registered unit for persistence is the early-map topic; keyword units show no detectable persistence on held-out papers (b +0.16 [−0.34, +0.66], unregistered). The empirical-Bayes tie rule was changed after outcome-free outputs were seen (disclosed). On the producer side (outcome-free; post hoc; D/E), the keywords with the largest raw growth in one year are among the top growers the next year less often than the base rate (16.0% against 20.6%; lift −0.045 [−0.075, −0.017]; Holm 0.018; 2,491 keyword-years), while those with the largest shrunk growth persist (40.3% against 22.1%; Holm 0.006). The raw reversal is confined to a one-year horizon and to keywords with 10–19 users, and on topic maps the raw lift is not negative (early map +0.04, Holm 0.25; all-years map +0.11). Last year's fastest-growing keywords are thus partly noise, as topics' past acceptance edges are.

Link-removed text. URLs and availability sentences were removed from the abstracts of decided 2019–2026 discovery papers, which changed the text of 22.4% of accepted and 8.5% of rejected papers, and the abstracts were re-embedded. The declared far-jump and field-distance families were rerun with frozen code; their robustness variants were not: E1 −5.44 → −5.39 (Holm 2.0×10^{-8}), E4 +0.55 → +0.53 (Holm 0.009), P1b −3.27 → −3.51 (Holm 0.0004), P7 +0.39 → +0.46 (Holm 0.035; consistent, not upgraded), P2 +0.68 → +0.98 (n.s.). The check does not cover rewritten results or polished wording, labels, topics or titles, and no link-removed text exists for held-out papers.

Year	n	Colon	D1	D2	D4	D3
2017	384	19.0	14.3	10.9	7.6	6.5
2018	774	22.4	16.8	13.8	8.8	7.9
2019	1,255	25.6	19.8	17.6	11.8	10.7
2020	2,071	26.4	21.4	18.8	14.6	13.8
2021	2,364	27.8	21.1	18.7	14.1	13.2
2022	2,649	31.4	24.6	21.9	17.3	16.0
2023	3,872	32.4	25.9	23.2	19.0	17.6
2024	5,766	40.8	32.9	30.2	25.3	23.9
2025	8,960	49.5	40.7	37.6	31.2	29.3
2026	15,420	59.7	48.8	44.6	38.8	37.0

Table 16: Share of titles (%) with any colon and under definitions D1–D4, by year. Outcome-free (E*, re-verified).

What this means for the test. None of these modules shows a premium that changes with how many others follow. Interactions with topic heat are null, and the edges of both topics and keywords regress. The FM premium does not detectably change with adoption, which fits rewarded imitation and efficient choice alike (Section 2); at the package level the gate cannot separate the routes. What acceptance does go with, beyond the record, is who writes (team size, experience), which is why Section 6 orders the evidence by how well it holds the writer fixed.

G Title Style

Surface form is where an imitation-like spread is plainest. This appendix gives the definitions and yearly shares behind Section 4 (E*, earlier exploratory, share re-verified for this paper).

Definitions. The prefix is the text before a title's first colon, and the subtitle after it must be non-empty. D1 (broad): the prefix has 1–3 whitespace tokens. D2 (name-like): D1, every prefix token starts with an upper-case letter or a digit, and the first token is not a generic lead word (for example Towards, Beyond, Rethinking, Learning, Understanding, Enhancing, On, How, What, Position). D3 (coined single token): one token with at least two capitals, or a digit and a capital (as in BERT, LoRA, GraphSAGE, Llama-2). D4 (one name-like token): D2 restricted to exactly one token. Titles come from the raw discovery file (43,515 rows with a non-empty title), so yearly counts differ slightly from the 43,480-paper corpus. No outcome column is read.

Shares. Under every definition the share of titles that open with a short coined name and a colon

rose roughly fourfold, from 7–14% (2017) to 37–49% (2026), by 3.4 times (D1) to 5.7 times (D3), and it rises every year after 2021 (Table 16). The level depends on the definition by about ±4 pp in 2017 and ±6 pp in 2026; the trend does not. The 2017 share rests on 384 titles (sampling SE about 1.6 pp at 11%). Whether accepted papers' titles were edited after the decision was not checked.

Acceptance (D*). In an earlier exploratory analysis, accepted papers carried branded titles at the submitted rate (ratio 1.02; odds ratio 1.00). That analysis is earlier exploratory and was not re-run. Two further earlier observations (on returning authors and on title wording) were not re-verified and are not reported.

What this means for the test. The title form spread roughly fourfold with no detectable premium at the gate in the earlier analysis: an imitation-like spread in form, consistent with bandwagons that go unrewarded (Abrahamson and Rosenkopf, 1993).

H Size Calibration

A registered test is only as good as its size and power. This appendix reports both, including a size failure in the entry tests on which the reversal of the herding prediction rests.

Dry runs. Before the freeze, two dry runs of 12,400 simulated jobs each (same seeds) measured size, false support under the full decision rule, and power. The entry nulls were calibrated on the corpus's own topic-year structure; the gate nulls used fixed parameters (base rate 0.32, two variance settings) plus a within-year permutation of the real acceptance outcomes; power grids ran from 0.5 to 6 times the smallest effect of interest (SESOI).

The size failure. Under an entry null whose structure follows the 2017–2020 map, and so cuts across the all-years topics used for clustering, the main tests of H2b and H2c rejected 12.6% and 10.0% of the time at nominal 5%; all other sizes were 3.8–7.2%. As registered, their confirmatory p-values came from a wild cluster restricted bootstrap (Rademacher weights by topic, null imposed, 9,999 draws; Cameron et al., 2008). The validation run showed that the bootstrap does not fix the failure (12.4% and 9.6%); we report it rather than hide it. Two-way clustering by all-years and early-map topic (S11) had sizes of 4.4–7.6% in every null (7.6% and 7.0% in the cross-map null). Under the cross-map null, some entry hypothesis was supported in 6.4% of replications (2.0% under the all-years null), some directional claim was made in 11.2% (4.6%), and per-hypothesis false support was at most 3.2%. The registered rule offers no extra protection against such structure: all main-specification false positives also passed the robustness and hindsight requirements.

Test	SESOI	Disc.	Held-out	Est. / power
H1d	–	–	12.0	−7.98
H2a	2	3.6	–	−0.73
H2b / H2b′	2	1.7	2.8	+3.71
H2c / H2c′	1	1.1	2.4	−1.70
H3a	1	4.1 (2.8)	4.8	54% (88%)[p]
H3b	1	2.4 (2.1)	–	−0.36
H4-FL level	–	–	14.5	−11.3
E1 far jump	–	–	4.9	power 0.89
P1b	–	–	5.4	power 0.47
P7	–	–	≈1.0–1.1[d]	+0.39
Teams H1	–	–	≈4.5[d]	+3.42
A1	–	–	≈4.3[d]	+2.67
PH6 D	–	3.6	7.4	−5.7

Table 17: Smallest effects of interest (SESOI, ±) and MDE80 in pp (H3a, H3b per log unit). Disc.: frozen discovery dry run (high-variance setting; low-variance in parentheses). Held-out: registered, simulated at each test's Holm first step; [d] derived post hoc (2.49×SE). PH6: discovery MDE80 as reported with its discovery estimate, which is not a test. Est.: discovery estimate, for comparison; power at the discovery effect for E1 and P1b. [p] H3a power at three times the SESOI in the discovery dry run.

Why the entry reversal survives it. The reversed estimates are large relative to the simulated inflation: $t = 6.7$ and -4.4, against t-inflation factors of 1.31 (H2b) and 1.22 (H2c) in the one cross-map scenario. Dividing by these factors still gives Holm p of 4×10^{-6} and 1×10^{-3}, and S11 agrees (Holm p 4×10^{-8} and 2×10^{-4}). The factors are a pre-declared single-scenario bracket, not a correction; how strong cross-topic structure is in the real data is unknown, so these p-values and their intervals remain approximate. On held-out papers, the simulated one-sided null sizes were 4.5% (H1d), 2.0% (H2b′), 2.5% (H2c′), 5.0% (H3a) and 4.0% (H4-FL).

Power and reading rules. Table 17 collects the minimum detectable effects. Non-supported tests below their MDE80 are uninformative, not evidence of no association. The gate's crowding test therefore cannot tell effects below about 4–5 pp

per log unit from zero, so "no detectable premium" is the strongest statement it supports. Supported held-out tests whose MDE80 exceeds the discovery estimate confirm direction, not size. The federated-learning level test could easily miss a gap of the discovery size (MDE80 14.5 against -11.3). The post hoc area tests were conservative (permutation rejection 0.5–1.4% at nominal 5%).

What this means for the test. The size failure is disclosed, not corrected, and it bears on the entry tests; their reversal is large relative to it and held on held-out papers, whose simulated sizes were at or below 5%. The absence of a detectable crowd premium is a statement about effects above roughly 5 pp per log unit, not about smaller ones.